

Section : Common

Q.1

If $f(x) = a \log_e |x| + bx^2 + x$ has critical points at $x = -2$ and $x = 1$, then

1. $a + 2b = 0$
2. $a - 2b = 0$
3. $a + 4b = 0$
4. $a - 4b = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Match List-I with List-II

List-(I)

(Differential equation)

(A) $\frac{d^3y}{dx^3} + y^2 + e^{dy/dx} = 0$

(B) $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} + 1 = 0$

(C) $2x^2 \frac{d^2y}{dx^2} - 3\left(\frac{dy}{dx}\right)^2 + y = 0$

(D) $\frac{d^3y}{dx^3} + 2\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$

List-(II)

(Order and Degree)

(I) order = 3, degree = 1

(II) order = 3, degree not defined

(III) order = 2, degree = 3

(IV) order = 2, degree = 1

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.3

The function $f(x) = x^2 - 4x + 6$ is

(A) Strictly decreasing on $(-\infty, 2) \cup (2, \infty)$

(B) Strictly increasing on $(2, \infty)$

(C) Strictly increasing on $(-\infty, \infty)$

(D) Strictly decreasing on $(-\infty, 2)$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (B) and (D) only
3. (A), (B) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.4

If $x = 4t$ and $y = \frac{4}{t}$, then $\frac{d^2y}{dx^2}$ is

1. $-\frac{1}{t^2}$
2. $\frac{2}{t^3}$
3. $\frac{1}{2t^3}$
4. $\frac{8}{t^3}$

Options 1. 1

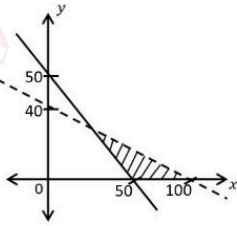
2. 2

3. 3

4. 4

Q.5

The constraints of the given shaded feasible region below of an L.P.P., for non-negative variable constraints x and y are



1. $x + y \leq 50, 2x + 5y \leq 200$

2. $x + y \geq 50, 2x + 5y \geq 200$

3. $x + y \leq 50, 2x + 5y \geq 200$

4. $x + y \geq 50, 2x + 5y < 200$

Options 1. 1

2. 2

3. 3

4. 4

Q.6

The number of all possible matrices of order 2×3 with each entry 0 or 1 is

1. 8

2. 16

3. 32

4. 64

Options 1. 1

2. 2

3. 3

4. 4

Q.7

If the optimal value of the objective function $z = px + y$ of an L.P.P occurs at two corner points $(2, 11)$ and $(4, 5)$ of its bounded feasible region, then its optimal value is

1. 15

2. 16

3. 18

4. 17

Options 1. 1

2. 2

3. 3

4. 4

Q.8

For $x \neq -1$, if $\int \frac{xe^x dx}{(1+x)^2} = \frac{ae^x}{(1+x)^b} + c$, where a, b are fixed numbers and c is the integration constant, then $a + b$ is equal to

1. 0

2. 1

3. 2

4. 3

Options 1. 1

2. 2

3. 3

4. 4

Q.9

The general solution of the differential equation $(1 + e^x)dy + ye^x dx = 0$, where $y > 0$, is

1. $y = C(1 + e^x)$, C is an arbitrary constant

2. $\frac{c}{e^x} = \frac{y}{1 + e^x}$, C is an arbitrary constant

3. $C = y(1 + e^x)$, C is an arbitrary constant

4. $y = \frac{Ce^x}{1 + e^x}$, C is an arbitrary constant

Options 1. 1

2. 2

3. 3

Q.10

Let $A = [a_{ij}]_{3 \times 3}$ such that $|A| = -5$. Then the value of $\det(5A)$ is equal to

1. 125
2. -125
3. 625
4. -625

Options 1. 1

2. 2
3. 3
4. 4

Q.11

If A is a square matrix such that $A^2 = A$ and I is the identity matrix of same order as A , then the matrix $(2I + A)^3 - 19A - 3I$ is equal to

1. $5I$
2. $5A$
3. $7I$
4. $7A$

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Match List-I with List-II

List-I

List-II

(A) The value of $\int_0^4 |x| dx$ is (I) 3

(B) The value of $\int_{-2}^2 |x| dx$ is (II) -1

(C) The value of $\int_0^3 [x] dx$ is (III) 8

(D) The value of $\int_{-1}^1 [x] dx$ is (IV) 4

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.13

The area of the region bounded by parabola $x^2 = 4y$, straight line $x = 2$ and x -axis, is

1. $\frac{4}{3}$ Sq. units
2. $\frac{2}{3}$ Sq. units
3. 1 Sq. unit
4. $\frac{5}{3}$ Sq. units

Options 1. 1

2. 2
3. 3
4. 4

Q.14

If A is an invertible matrix of order 2, then $\det((\text{adj } A)^{-1})$ is equal to

1. 0
2. 1
3. $\det A$
4. $\frac{1}{\det A}$

Options 1. 1

2. 2

3. 3
4. 4

Q.15

Let x denote the number of doublets in three throws of a pair of dice with the following probability distribution.

x 0 1 2 3

$$P(x) \quad \frac{25}{72}k \quad \frac{15}{72}k \quad \frac{3}{72}k \quad \frac{1}{360}k$$

If value of k is equal to $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to

1. 8
2. 19
3. 16
4. 18

Options 1. 1

2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16

The probability of a man hitting a target is $\frac{1}{2}$.

How many times must he fire so that the probability of hitting the target at least once is more than 90%?

1. 3
2. 4
3. 5
4. 6

Options 1. 1

2. 2
3. 3
4. 4

Q.17

A boat takes 1 hour 30 minutes less to travel 36 km downstream than to travel the same distance upstream.

If the speed of the boat in still water is 10 km/hr, then the speed of the stream is:

1. 2 km/hr
2. 3 km/hr
3. 3.5 km/hr
4. 4 km/hr

Options 1. 1

2. 2
3. 3
4. 4

Q.18

Year (t)	2010	2011	2012	2013	2014	2015	2016
Sales (in Rs. crores) (y)	6	8	9	11	13	17	20

In reference to the above data, which of the following statements are correct?

(where $x = t - 2013$)

(A) If the equation of the straight line trend is $y = 12 + 2.29x$, then the trend value for the year 2017 is 21.16.

(B) If the equation of the straight line trend is $y = 12 + 2.29x$, then the trend value for the year 2013 is 11.

(C) If the equation of the straight line trend is $y = 12 + 2.29x$, then the trend value for the year 2015 is 17.

(D) If $(t_1, y_1), (t_2, y_2), \dots, (t_n, y_n)$ denote the time series and y_t are the trend values of y , then $\sum (y - y_t) = 0$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (A), (B) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.19

A tank has two outlet pipes A and B, which together take 6 hours to empty a full tank when opened simultaneously.

The tank was initially half-full and both outlets were opened.

After one hour, an inlet pipe was also opened.

If the inlet alone can fill the empty tank in 4 hours,
how much time will it now take to fill the tank completely?

1. 9 hrs
2. 8 hrs
3. 6 hrs
4. 11 hrs

Options 1. 1
2. 2
3. 3
4. 4

Q.20 Which of the following are correct?

- (A) If A and B are symmetric matrices such that $AB = BA$, then AB is symmetric.
(B) If A and B are symmetric matrices of the same order, then $(A + B)$ is a symmetric matrix.
(C) If A and B are symmetric matrices of the same order, then $(AB - BA)$ is a symmetric matrix.
(D) If A and B are symmetric matrices of the same order, then $(AB + BA)$ is a skew-symmetric matrix.

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.21 In a game of billiards, A can give B 15 points in 60 and A can give C 20 points in 60.

How many points can B give C in a game of 90?

1. 10 points
2. 15 points
3. 20 points
4. 12 points

Options 1. 1
2. 2
3. 3
4. 4

Q.22 Let A be a square matrix of order 2 such that

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then A is:

1. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
2. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
3. $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
4. $\begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$

Options 1. 1
2. 2
3. 3
4. 4

Q.23 Let the corner points of the bounded feasible region of the linear programming problem

(LPP) $Z = ax + by$ be $(0, 0)$, $(2, 0)$, $\left(\frac{20}{19}, \frac{45}{19}\right)$ and $(0, 3)$.

If the optimal value of Z occurs at both points $(2, 0)$ and $\left(\frac{20}{19}, \frac{45}{19}\right)$,

then the relation between a and b is:

1. $a = 3b$
2. $5a = 2b$
3. $2a = 5b$
4. $3a = 2b$

Options 1. 1
2. 2
3. 3
4. 4

Q.24 Under which of the following conditions the Poisson distribution is the limiting case of, binomial distribution:

- (A) The number of trials is indefinitely large.

- (B) The probability of success for each trial is indefinitely small.
 (C) The product of the number of trials and the probability of success for each trial is finite.
 (D) The probability of success for each trial is indefinitely large.
 Choose the correct answer from the options given below:
 1. (A), (B) and (C) only
 2. (A) and (B) only
 3. (B) and (C) only
 4. (A), (C) and (D) only

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.25

Mr. X invested Rs. 4, 00, 000 in shares for 5 years.
 The value of this investment was Rs. 4, 50, 000 at the end of the second year,
 Rs. 4, 90, 000 at the end of the third year and on maturity,
 the final value stood at Rs. 6, 00, 000.
 The compound annual growth rate (CAGR) of this investment is:
 [Given that $(1.5)^{1/5} = 1.084$]
 1. 8.4%
 2. 10.8%
 3. 9.084%
 4. 11%

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.26

Let $A = [a_{ij}]$ be a square matrix of order 3
 with each entry either 0 or 1.
 Then the number of all such possible matrices is:
 1. 256
 2. 81
 3. 243
 4. 512

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.27

Pure milk costs Rs. 75 per litre.
 A milkman adds water to 28 litres of pure milk
 and sells the mixture at Rs. 60 per litre.
 How many litres of water does he add?
 1. 5
 2. 6
 3. 7
 4. 6.5

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.28

At what rate of interest will the present value of a perpetuity of
 Rs. 1000 payable at the end of every six months be Rs. 20000?
 1. 10%
 2. 12%
 3. 8%
 4. 11%

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.29

For predicting the straight line trend in the sales of cars (in thousands)
 on the basis of 5 consecutive years' data, the company uses a
 3-year moving averages method.
 If the sales of cars for respective years are 15, 24, 18, 33 and 42,
 then which of the following averages will not be computed?
 1. 19
 2. 25
 3. 32
 4. 31

Options 1. 1
 2. 2
 3. 3
 4. 4

Q.30 In a game, a man wins Rs. 5 for getting a number greater than 4 and loses Rs. 1 otherwise, when a fair die is thrown. The man decides to throw the die thrice but quits as soon as he gets a number greater than 4. The expected value of the amount (in Rs.) he wins or loses is:

1. $\frac{14}{9}$
2. $\frac{19}{9}$
3. $\frac{22}{9}$
4. $\frac{25}{9}$

Options 1. 1
2. 2
3. 3
4. 4

Q.31 The inequality $\frac{5x-2}{3} - \frac{7x-3}{5} > \frac{x}{4}$ holds when

1. $x \in (-4, \infty)$
2. $x \in (4, \infty)$
3. $x \in (-\infty, 2]$
4. $x \in (-\infty, 4]$

Options 1. 1
2. 2
3. 3
4. 4

Q.32 If the function $f(x) = 2x^3 + 9x^2 + 12x - 1$ is given, then $f(x)$ has

1. Local minima at $x = -1$
2. Local maxima at $x = \frac{3}{2}$
3. Neither maxima nor minima at $x = -2$
4. Minimum value of $f(x)$ is $\frac{3}{2}$

Options 1. 1
2. 2
3. 3
4. 4

Q.33 The remainder when 2^{340} is divided by 341 is:

1. 0
2. 1
3. -1
4. 2

Options 1. 1
2. 2
3. 3
4. 4

Q.34 Let $\begin{vmatrix} 3 & y \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$ and x, y are natural numbers, then the number of solutions for the system is:

1. 1
2. 2
3. 4
4. 8

Options 1. 1
2. 2
3. 3
4. 4

Q.35 The rise in prices before festivals is an example of

1. Long term trend
2. Cyclical trend
3. Seasonal trend
4. Irregular trend

Options 1. 1
2. 2
3. 3
4. 4

Q.36

- Which of the following are correct about equated monthly installments (EMI)?
- (A) The EMI depends on principal borrowed, rate of interest and tenure of the loan.
(B) It is a fixed amount made by borrower to the lender every month.
(C) The interest remains constant for every EMI in reducing balance method.
(D) As we pay off our loan, the outstanding principal amount decreases with every EMI in reducing balance method.
- Choose the correct answer from the options given below:
1. (A), (B) and (C) only
 2. (A), (B) and (D) only

3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.37

A 95% confidence interval states that the population mean is greater than 152 and less than 160.

If $\sigma = 15$ and $Z_{0.025} = 1.96$, then what sample size was used in the study?

1. 50
2. 48
3. 64
4. 54

Options 1. 1
2. 2
3. 3
4. 4

Q.38

List 1

List 2

(A) If A is a non singular matrix of order n, then $|A(\text{adj} A)|$ is equal to (I) $|A|^{n-1}$

(B) If A is a non singular matrix of order n, then $|\text{adj}(\text{adj} A)|$ is equal to (II) $|A|^{n-2} A$

(C) If A is a non singular matrix of order n, then $\text{adj}(\text{adj} A)$ is equal to (III) $|A|^n$

(D) If A is a non singular matrix of order n, then $|(\text{adj} A)|$ is equal to (IV) $|A|^{(n-1)^2}$

Match List-I with List-II

Choose the correct answer from the options given below:

1. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
2. (A) – (II), (B) – (I), (C) – (III), (D) – (IV)
3. (A) – (III), (B) – (IV), (C) – (II), (D) – (I)
4. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.39

If $I = \int \frac{x^4 + x^2 + 1}{x^2 - x + 1} dx = ax + \beta x^2 + \gamma x^3 + \delta$, where δ is the constant of integration,

then $(a + 2\beta + 3\gamma)$ equals

1. 0
2. 1
3. 2
4. 3

Options 1. 1
2. 2
3. 3
4. 4

Q.40

Let $y(x) = a(x+1)\log(x+1) + bx + 5$ be the solution of the differential equation $e^{\frac{dy}{dx}} = x + 1$, $y(0) = 5$, then the value of $(a + b)$ is:

- (1) 0
(2) 1
(3) 2
(4) -1

Options 1. 1
2. 2
3. 3
4. 4

Q.41 Which of the following is not the specification of the Sinking Fund?

1. It is a fixed term account.
2. It can be used in an emergency.
3. A fixed amount at regular intervals is deposited in the Sinking Fund.
4. It is set-up for a particular upcoming expense.

Options 1. 1
2. 2
3. 3
4. 4

Q.42

For the linear programming problem (LPP),

Maximize $Z = 7x + 9y$

subject to the constraints

- $x - y \leq -1$,
 $-x + y \leq 0$,
 $x, y \geq 0$.

Which of the following is correct?

1. The feasible region is bounded.
2. The optimal value of the objective function exists at the point (1, 1).
3. The optimal value of the objective function is 16.
4. No feasible region exists.

Options 1. 1
2. 2
3. 3
4. 4

Q.43

Which of the following are correct?

- (A) The function $f(x) = 3x + 12$ is increasing on $\mathbb{R}$.
- (B) The function $f(x) = e^{2x}$ is decreasing on $\mathbb{R}$.
- (C) The function $f(x) = x^2 - x + 1$ is neither increasing nor decreasing on $(-1, 1)$.
- (D) The function $f(x) = x^3 - 3x^2 + 4x$ is increasing on $\mathbb{R}$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (C) and (D) only
3. (C) and (D) only
4. (A), (B) and (C) only

Options 1. 1
2. 2
3. 3
4. 4

Q.44

Let us suppose that two independent random samples of sizes n_1 and n_2 have been drawn from the same normal population. Then the degree of freedom of the statistic t -distribution is

1. $n_1 + n_2 - 1$
2. $n_1 + n_2 - 2$
3. $n_1 + n_2 + 1$
4. $n_1 + n_2 + 2$

Options 1. 1
2. 2
3. 3
4. 4

Q.45

If X is a normal variate with mean 12 and standard deviation 4, then $P(X \geq 20)$ is:

[Given that $P(0 \leq Z \leq 2) = 0.4772$]

1. 0.9772
2. 0.0228
3. 0.00135
4. 0.7563

Options 1. 1
2. 2
3. 3
4. 4

Q.46 Which of the following is not correct about the Central Limit Theorem?

1. The sampling distribution of the sample mean approaches a bell-shaped curve as the sample size gets larger.
2. A sample size of 30 or more is considered to be sufficient to hold the Central Limit Theorem.
3. As the sample size becomes larger, the prediction of characteristics of the population becomes more accurate.
4. The sampling distribution of the sample mean approaches a Poisson distribution as the sample size gets larger.

Options 1. 1
2. 2
3. 3
4. 4

Q.47

A person has purchased a home for Rs. 10,00,000 with a down payment of Rs. 2,00,000.

He amortizes the balance at 9% per annum compounded monthly for 25 years.

Then the equal monthly instalment (EMI) is:

[Given that $\frac{(1.0075)^{300} - 1}{(0.0075)(1.0075)^{300}} = 119.1616$]

1. Rs. 8174.55
2. Rs. 6713.57
3. Rs. 7248.64
4. Rs. 7534.25

Options 1. 1
2. 2
3. 3
4. 4

Q.48

A motorbike costing Rs. 1,25,000 has a scrap value of Rs. 25,000.

If the annual depreciation charge is Rs. 12,500, then the useful life of the bike is (by using linear method):

1. 7 years
2. 8 years

3. 9 years
4. 10 years

Options 1. 1
2. 2
3. 3
4. 4

Q.49

The value of the integral $I = \int_0^1 \frac{1}{\sqrt{x^2 + 2x + 3}} dx$ is:

1. $\log \left(\frac{2 + \sqrt{6}}{1 - \sqrt{3}} \right)$
2. $\log \left(\frac{2 + \sqrt{6}}{1 + \sqrt{3}} \right)$
3. $\log \left(\frac{5 + \sqrt{6}}{1 + \sqrt{3}} \right)$
4. $\log(5 + \sqrt{6})$

Options 1. 1
2. 2
3. 3
4. 4

Q.50 The level of production where the revenue from sales is equal to the cost of production and marketing is known as

1. the marginal cost point.
2. the fixed cost point
3. the break even point.
4. the fixed profit point.

Options 1. 1
2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51

If α, β and γ are angle of inclinations of a line with x, y and z axes respectively, then the value of $2(\cos 2\alpha + \cos 2\beta + \cos 2\gamma)$ is

1. 1
2. -1
3. 2
4. -2

Options 1. 1

2. 2
3. 3
4. 4

Q.52

The area (in sq. units) bounded by the parabola $y^2 = 16x$ and its latus rectum is

1. 64 Sq. units
2. $\frac{32}{3}$ Sq. units
3. $\frac{64}{3}$ Sq. units
4. $\frac{128}{3}$ Sq. units

Options 1. 1
2. 2
3. 3
4. 4

Q.53

The position vector of a point which divides the line joining the points with position vectors $(\vec{a} - 2\vec{b})$ and $(2\vec{a} + \vec{b})$ externally in the ratio 2 : 1, is

1. $3\vec{a} + 4\vec{b}$
2. $-3\vec{a} + 4\vec{b}$
3. $3\vec{a} - 4\vec{b}$
4. $\frac{5}{3}\vec{a}$

Options 1. 1

2. 2
3. 3
4. 4

Q.54

The maximum value of $f(x) = \frac{1}{4x^2 + 2x + 1}$ is

1. $\frac{3}{4}$
2. $\frac{4}{3}$
3. $\frac{2}{3}$
4. $\frac{1}{3}$

Options 1. 1
2. 2
3. 3
4. 4

Q.55

If $\vec{p}$ and $\vec{q}$ are two unit vectors such that $|\vec{p} + \vec{q}| = \sqrt{2}$, then which of the following are correct?

- (A) $|\vec{p}| = |\vec{q}| = 1$
- (B) $\vec{p}$ and $\vec{q}$ are orthogonal vectors
- (C) $\vec{p}$ and $\vec{q}$ are collinear vectors
- (D) $(4\vec{p} - \vec{q}) \cdot (2\vec{p} + \vec{q}) = 7$

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) and (C) only
3. (A), (C) and (D) only
4. (A), (B) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.56

If $A = \begin{bmatrix} -2 \\ -1 \\ -4 \end{bmatrix}$, $B = [-1 \ 2 \ 3]$, then the value of $A'B'$ is

1. $\begin{bmatrix} 2 & -4 & -6 \\ 1 & -2 & -3 \\ 4 & -8 & -12 \end{bmatrix}$
2. $\begin{bmatrix} 2 & 1 & 4 \\ -4 & -2 & -8 \\ -6 & -3 & -12 \end{bmatrix}$
3. $[12]$
4. $[-12]$

Options 1. 1
2. 2
3. 3
4. 4

Q.57

The straight line $\frac{x+3}{3} = \frac{y+2}{4} = \frac{z+1}{0}$ is

1. Parallel to x -axis
2. Parallel to y -axis
3. Parallel to z -axis
4. Perpendicular to z -axis

Options 1. 1
2. 2
3. 3
4. 4

Q.58

If $\int e^x \left(\frac{x-1}{(x+1)^3} \right) dx = \frac{Ae^x}{(x+1)^B} + C$, where C is constant of integration, then which of the following are correct?

- (A) $A = -1$
- (B) $A = 1$
- (C) $B = 3$
- (D) $B = 2$

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (B) only
3. (B) and (D) only

4. (A) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.59

If $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ then

Match List-I with List-II

List-I

(A) Projection of $\vec{a}$ on $\vec{b}$ is

(B) $\vec{a} \times \vec{b}$ is

(C) Unit vector along $\vec{a} + \vec{b}$ is

(D) Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is

List-II

(I) $-7\hat{i} + 4\hat{j} + 6\hat{k}$

(II) $\frac{1}{\sqrt{101}}(-7\hat{i} + 4\hat{j} + 6\hat{k})$

(III) $\frac{5}{3}$

(IV) $\frac{1}{\sqrt{33}}(4\hat{i} + \hat{j} + 4\hat{k})$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)

2. (A) – (III), (B) – (I), (C) – (II), (D) – (IV)

3. (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

4. (A) – (I), (B) – (IV), (C) – (II), (D) – (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.60

Which of the following functions are increasing on $x \in \left(0, \frac{\pi}{2}\right)$?

(A) $f(x) = \sin x$

(B) $f(x) = \cos x$

(C) $f(x) = \tan x$

(D) $f(x) = \cos 3x$

Choose the correct answer from the options given below:

1. (A) and (D) only

2. (A) and (C) only

3. (B) and (C) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.61

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 7$, $|\vec{b}| = 3$ and $|\vec{c}| = 5$, then angle between $\vec{b}$ and $\vec{c}$ is

1. $\frac{2\pi}{3}$

2. $\frac{\pi}{6}$

3. $\frac{5\pi}{6}$

4. $\frac{\pi}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.62

The value of $\begin{vmatrix} 1 & x & y \\ 1 & x+y & y \\ 1 & x & x+y \end{vmatrix}$ is

1. 0

2. $-xy$

3. xy

4. x^2y^2

- Options 1. 1
2. 2
3. 3
4. 4

Q.63

If a computer code is correctly programmed, it gives 90\% acceptable results. But if it is not correctly programmed, it gives only 40\% acceptable results. From previous experience, it is observed that only 80\% of codes are correctly programmed. If after a certain programming, the code gives 2 acceptable results, then the approximate probability that the code is correctly programmed is

1. $\frac{81}{85}$
2. $\frac{9}{10}$
3. $\frac{17}{20}$
4. $\frac{13}{20}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.64

Let $A = \{1, 2, 3\}$, then the possible equivalence relations on A are:

- (A) $\{(1, 1), (2, 2), (3, 3)\}$
- (B) $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$
- (C) $\{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3)\}$
- (D) $\{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (B), (C) and (D) only
3. (C) and (D) only
4. (A), (B) and (C) only

- Options 1. 1
2. 2
3. 3
4. 4

Q.65

For $x \in (0, \frac{\pi}{2})$, $\int \frac{1}{\sin^2 x + \sin 2x} dx$ is equal to

1. $\frac{1}{2} \log \left| \frac{\tan x + 2}{\tan x - 2} \right| + C$, where C is constant of integration
2. $\frac{1}{2} \log \left| \frac{\tan x}{\tan x + 2} \right| + C$, where C is constant of integration
3. $\frac{1}{2} \log \left| \frac{\tan x + 2}{\tan x} \right| + C$, where C is constant of integration
4. $\frac{1}{2} \log \left| \frac{\tan x + 1}{\tan x + 2} \right| + C$, where C is constant of integration

- Options 1. 1
2. 2
3. 3
4. 4

Q.66 **Match List-I with List-II**

An urn contains 4 white and 3 red balls. In a random draw of three balls, the probability of

List-I **List-II**

(A) No red ball is (I) $\frac{12}{35}$

(B) Only 1 red ball is (II) $\frac{1}{35}$

(C) Exactly 2 red balls is (III) $\frac{4}{35}$

(D) No white ball is (IV) $\frac{18}{35}$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)
2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
3. (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
4. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.67

If $A = \begin{bmatrix} 0 & l & -3 \\ -2 & 0 & 1 \\ m & -1 & 0 \end{bmatrix}$ is a skew symmetric matrix, then

1. $l = 3, m = 2$

2. $l = -3, m = 2$

3. $l = 2, m = 3$

4. $l = 2, m = -3$

Options 1. 1

2. 2

3. 3

4. 4

Q.68

The maximum value of the objective function $z = 2x + 3y$ of an L.P.P. subjected to the constraints

$x - y \leq 1, x + y \leq 3, x, y \geq 0$, is

1. 11

2. 9

3. 7

4. 5

Options 1. 1

2. 2

3. 3

4. 4

Q.69

Consider the lines $l_1 : \frac{x-1}{0} = \frac{y-1}{1} = \frac{2-z}{1}$ and $l_2 : \frac{x}{2} = \frac{y}{0} = \frac{2z-1}{4}$, then which of the following are correct?

(A) Direction Ratios of $l_1 = \langle 0, 1, 1 \rangle$

(B) Direction Ratios of $l_2 = \langle 2, 0, 2 \rangle$

(C) Angle between l_1 and $l_2 = \frac{\pi}{3}$

(D) Angle between l_1 and $l_2 = \frac{2\pi}{3}$

Choose the correct answer from the options given below:

1. (B) and (D) only

2. (A) and (C) only

3. (A), (B) and (C) only

4. (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.70

If the area of an equilateral triangle is increasing at the rate of $4\sqrt{3} \text{ cm}^2/\text{sec}$, then the rate of increase of its perimeter when the side is 4 cm, is

1. 2 cm/sec

2. 3 cm/sec

3. 4 cm/sec

4. 6 cm/sec

Options 1. 1

2. 2

3. 3

4. 4

Q.71

If A is a square matrix, then $(A^T - A)$ is-

1. a null matrix

2. an identity matrix

3. a symmetric matrix

4. a skew-symmetric matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.72

If the function defined by $f(x) = \begin{cases} kx^2 + 1, & \text{if } x \leq 1 \\ 2, & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$, then k is equal to

1. 2

2. 3

3. -1

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.73

For $x \in (0, \frac{\pi}{2})$, $\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$ is equal to

1. $\sin^{-1}(\sin 2x) + C$, C is an arbitrary constant

2. $\sin^{-1}(\cos x - \sin x) + C$, C is an arbitrary constant

3. $\sin^{-1}(\sin x - \cos x) + C$, C is an arbitrary constant

4. $\sin^{-1}(\sin x + \cos x) + C$, C is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.74

The area of a triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is given by the absolute value of

1. $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

2. $\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

3. $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 0 \\ x_3 & y_3 & 0 \end{vmatrix}$

4. $\Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 0 \\ x_3 & y_3 & 0 \end{vmatrix}$

Options 1. 1

2. 2

3. 3

4. 4

Q.75

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2, & x \geq 1 \\ x, & x < 1 \end{cases}$$
 is

1. Continuous but not differentiable at $x = 1$

2. Continuous and differentiable at $x = 1$

3. Neither continuous nor differentiable at $x = 1$

4. Continuous but not differentiable at $x = 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.76

Which of the following functions from $\mathbb{Z}$ to $\mathbb{Z}$ is a bijective function? (where $\mathbb{Z}$ is set of integers)

1. $f(x) = x^3$

2. $f(x) = 2x + 1$

3. $f(x) = x^2 + 1$

4. $f(x) = x + 2$

Options 1. 1

2. 2

3. 3

Q.77

If $2P(A) = P(B) = \frac{5}{13}$ and $P(A|B) = \frac{2}{5}$, then $P(A \cap B)$ is

1. $\frac{1}{13}$
2. $\frac{2}{13}$
3. $\frac{3}{13}$
4. $\frac{4}{13}$

Options 1. 1

2. 2
3. 3
4. 4

Q.78

Consider the differential equation $\frac{dy}{dx} + y \tan x = \sec x$, then which of the following statements are correct?

- (A) It is homogeneous
- (B) It has $\sec x$ as its integrating factor
- (C) Its general solution is $y \sec x = \tan x + c$, where c is an arbitrary constant.
- (D) Its degree is not defined

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (B) and (C) only
3. (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.79

For the principal value branch, the value of $\sin \left(\frac{\pi}{2} - \sin^{-1} \left(-\frac{\sqrt{3}}{2} \right) \right)$ is

1. $\frac{1}{\sqrt{2}}$
2. $\frac{\sqrt{3}}{2}$
3. $\frac{1}{2}$
4. $-\frac{\sqrt{3}}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.80

A linear programming problem is as follows:

Minimize $z = 2x + 3y$

Subject to the constraints $x \geq 3$, $x \leq 9$, $y \geq 0$, $x - y \geq 0$, $x + y \leq 14$.

The feasible region has 5 corner points including

1. (0, 0) and (9, 5)
2. (14, 0) and (9, 0)
3. (7, 7) and (3, 3)
4. (3, 6) and (9, 5)

Options 1. 1

2. 2
3. 3
4. 4

Q.81

The system of equations

$$x + y + z = 7$$

$$x + 2y + 3z = 5$$

$$x + 3y + \lambda z = \mu$$

has a unique solution, if

1. $\lambda = 5$ and $\mu = 7$
2. $\lambda \neq 1$ and $\mu = 5$
3. $\lambda \neq 5$ and μ is any real number
4. $\lambda \neq 1$ and $\mu \neq 5$

Options 1. 1

2. 2
3. 3
4. 4

Q.82 Match List-I with List-II

If $P(A) = \frac{3}{7}$, $P(B) = \frac{4}{7}$ and $P(A \cup B) = \frac{5}{7}$

List-I

List-II

(A) $P(A \cap B)$

(I) $\frac{2}{3}$

(B) $P(A|B)$

(I) $\frac{5}{7}$

(C) $P(B|A)$

(III) $\frac{1}{2}$

(D) $P(A' \cup B')$

(IV) $\frac{2}{7}$

Choose the correct answer from the options given below:

1. (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
2. (A) – (IV), (B) – (III), (C) – (II), (D) – (I)
3. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
4. (A) – (II), (B) – (I), (C) – (IV), (D) – (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.83

For $y \neq 0$, the particular solution of the differential equation $2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$ at the point $(1, 1)$ is

1. $\ln|y| + 2(e^{x/y} - e) = 0$
2. $\ln|y| - 2(e^{x/y} - e) = 0$
3. $\ln|y| + \frac{1}{2}(e^{x/y} - e) = 0$
4. $\ln|y| - \frac{1}{2}(e^{x/y} - e) = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.84

For $|x| < 1$, if $x = \cos\left(\frac{1}{a}\log y\right)$, then

1. $(1 - x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - a^2y = 0$
2. $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + a^2y = 0$
3. $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$
4. $(1 - x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} + a^2y = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.85

The area bounded by the curve $y = 4 + 3x - x^2$ and the x -axis is equal to

1. $\frac{125}{6}$ Sq. units
2. $\frac{125}{3}$ Sq. units
3. $\frac{125}{2}$ Sq. units

4. $\frac{121}{6}$ Sq. units

- Options
1. 1
 2. 2
 3. 3
 4. 4

